



Lab 4

Basic Logic Gate Implementation

Part 4

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Objectives

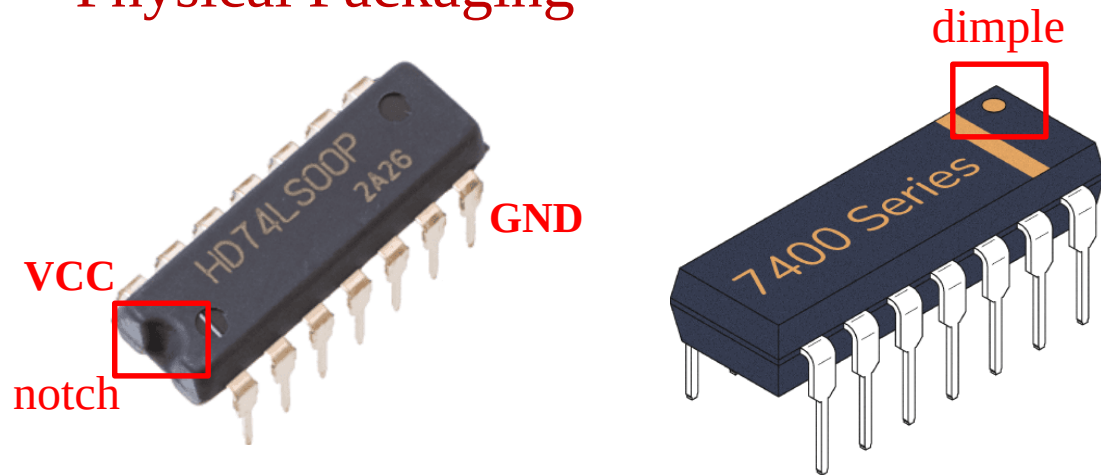
- Understand following components
 - Transistor-Transistor Logic (TTL) ICs 7400, 7404, 7408, 7432
- Implement **logic circuits** to verify the corresponding **truth tables**

TTL 7400 (NAND)

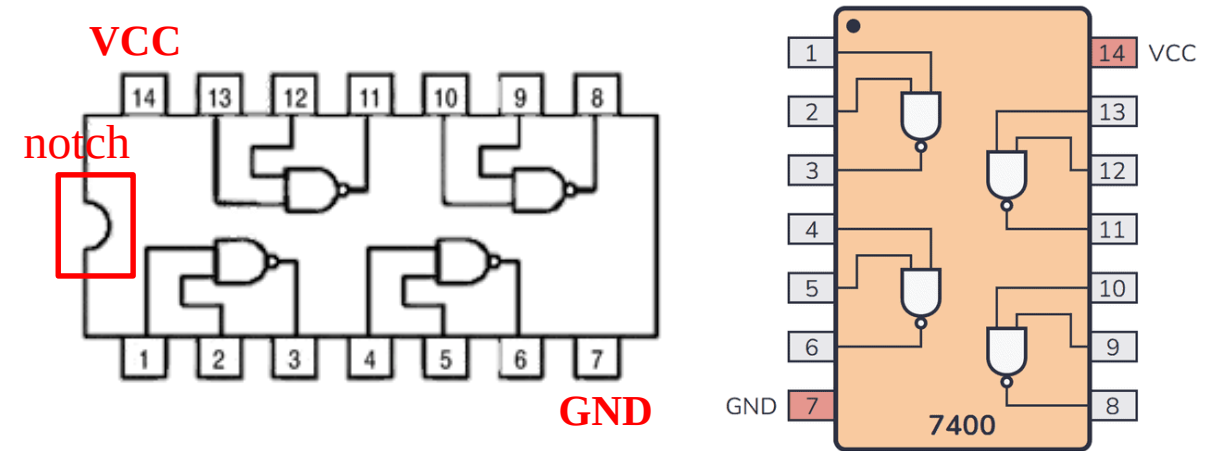
Figure Sources:

- <https://www.igeelectronics.com/products/g69b66b32e/1726810000000161177>
- <https://www.build-electronic-circuits.com/7400-series-integrated-circuits/74hc00-74ls00/>

Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

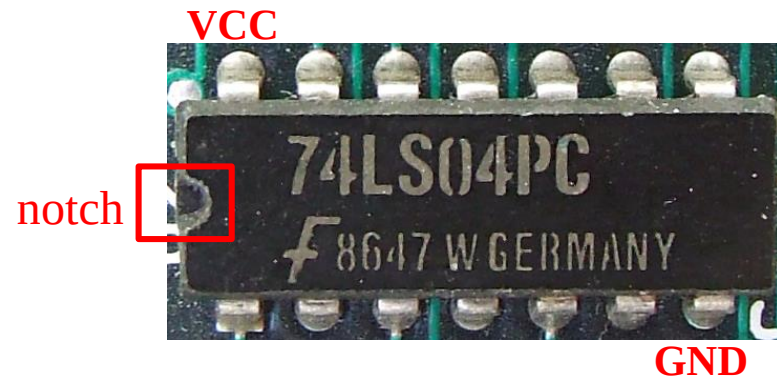
Input A	Input B	Output OUT
0	0	1
0	1	1
1	0	1
1	1	0

TTL 7404 (NOT)

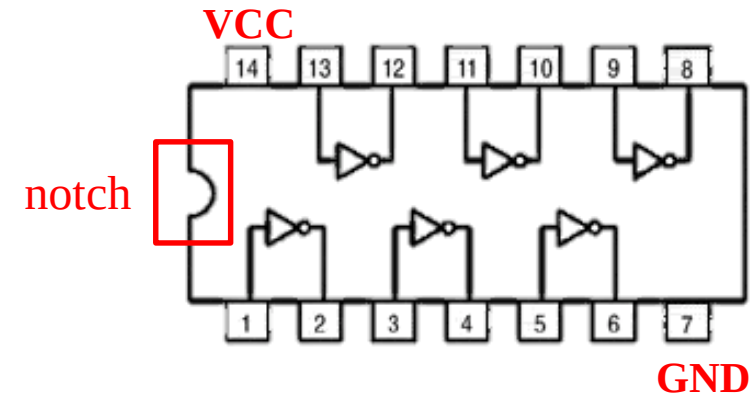
Figure Source:

1. https://pt.wikipedia.org/wiki/TTL_7404

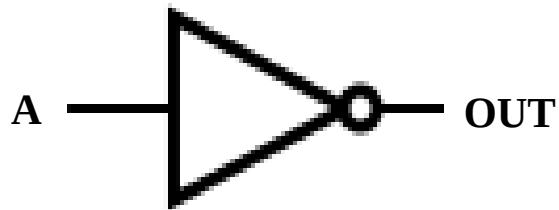
Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

Input A	Output OUT
0	1
1	0

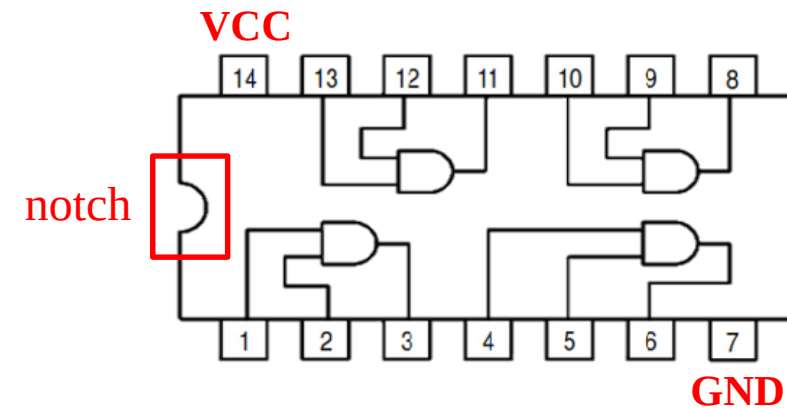
TTL 7408 (AND)

Figure Source:
1. https://pt.wikipedia.org/wiki/TTL_7408

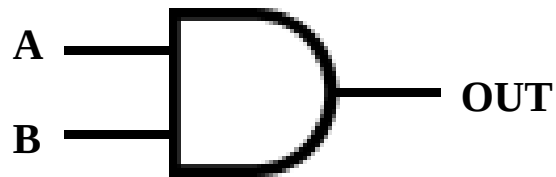
Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

Input A	Input B	Output OUT
0	0	0
0	1	0
1	0	0
1	1	1

TTL 7432 (OR)

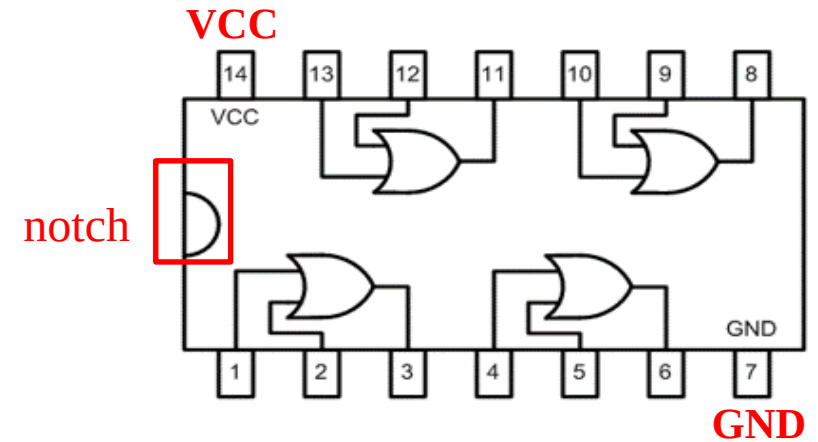
Figure Source:

<https://docs.cirkitdesigner.com/component/e39cedd2-1d38-4573-829f-9f133c89a3b8/7432-or-gate-ic>

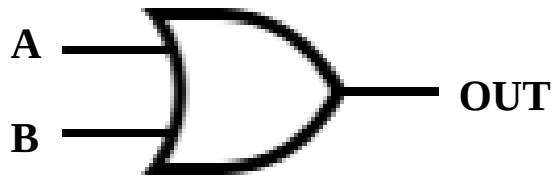
Physical Packaging



Internal Structure



Graphic Symbol



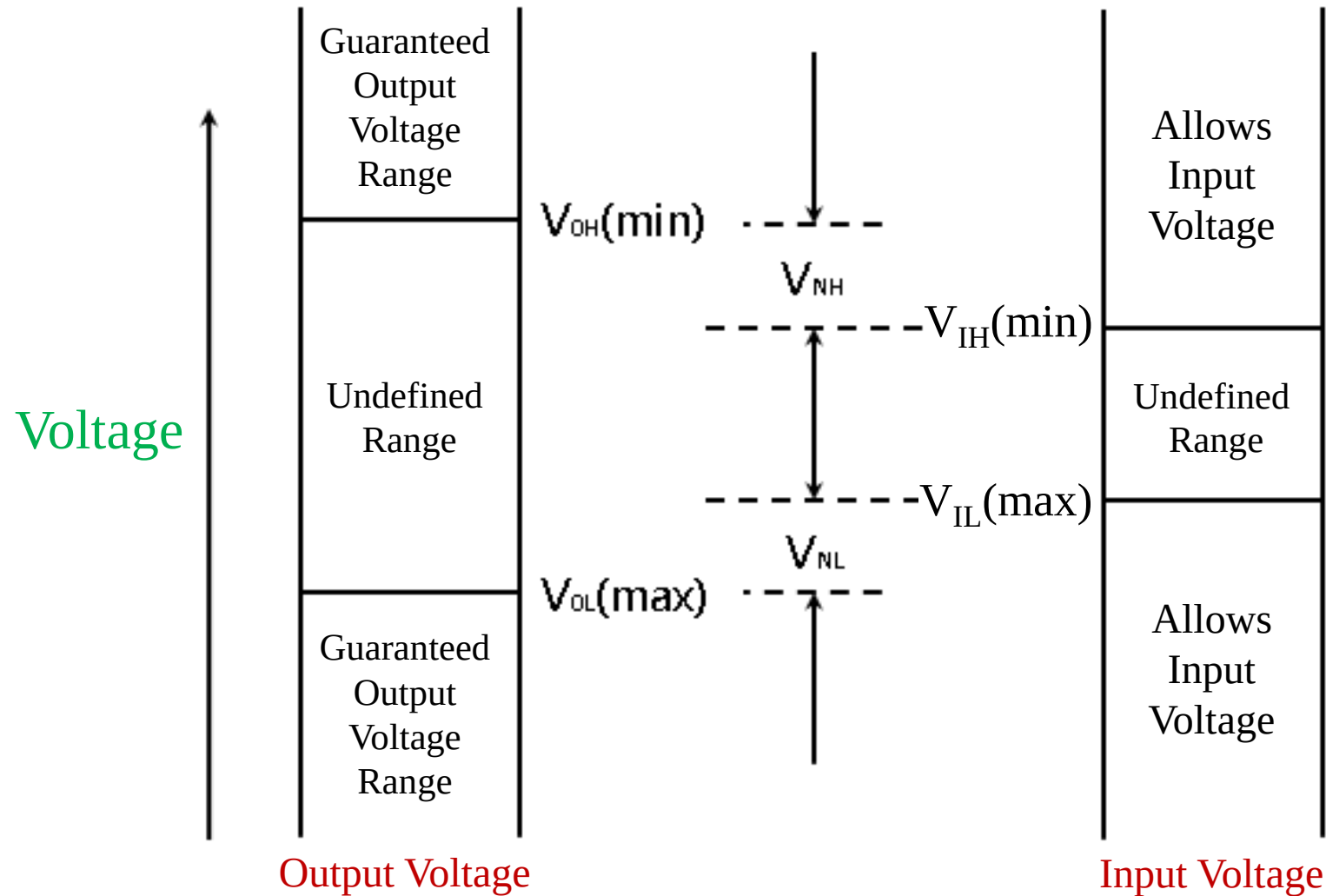
Truth Table

Input A	Input B	Output OUT
0	0	0
0	1	1
1	0	1
1	1	1

Number/Alphabet vs Logic State

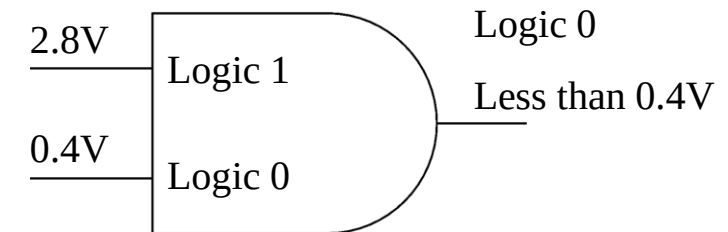
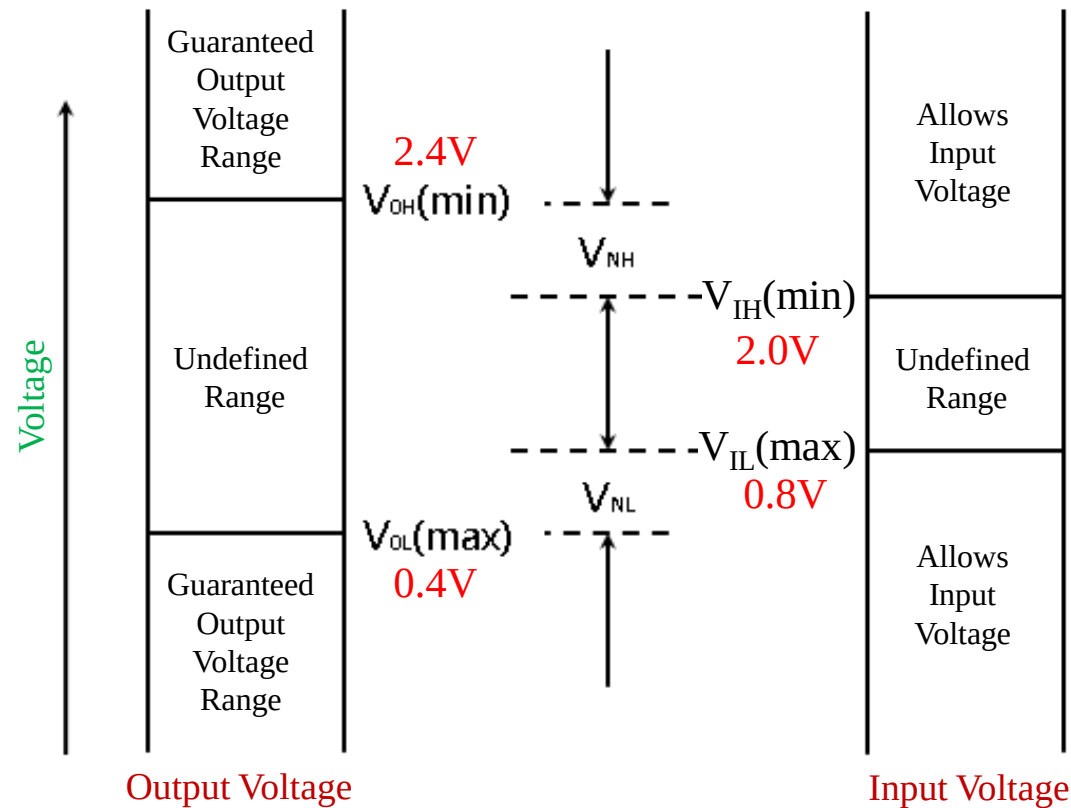
- 0 → Logic LOW (Ground)
- 1 → Logic HIGH (5V)
- x or X → Undefined
- z or Z → High Impedence or Floating
- When an input pin of a TTL IC is floating, it is typically interpreted as a logic HIGH (1).

Voltage Logic Levels



Example of Voltage Logic Levels --- Standard TTL

TTL Type	IC Number	$V_{OH}(\text{min})$	$V_{IH}(\text{min})$	$V_{OL}(\text{max})$	$V_{IL}(\text{max})$	Propagation Delay	Power Consumption
Standard	74xx	2.4V	2.0V	0.4V	0.8V	9 ns	10 mW



Comparison Table

TTL Type	IC Number	$V_{OH}(\text{min})$	$V_{IH}(\text{min})$	$V_{OL}(\text{max})$	$V_{IL}(\text{max})$	Propagation Delay	Power Consumption
Standard	74xx	2.4V	2.0V	0.4V	0.8V	9 ns	10 mW
Low Power	74Lxx	2.4V	2.0V	0.3V	0.8V	33 ns	1 mW
High-Speed	74Hxx	2.4V	2.0V	0.4V	0.8V	6 ns	23 mW
Schottky (high-speed with the same power consumption)	74Sxx	2.4V	2.0V	0.5V	0.8V	3 ns	23 mW
Low Power Schottky	74LSxx	2.4V	2.0V	0.5V	0.8V	9.5 ns	2 mW

LAB 4-1

- Input: A, B, C ; Output: $F(A, B, C) = \Sigma(1, 5, 6, 7)$
- Requirements
 - Complete the truth tables in voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the derived simplest Boolean expression, draw the circuit diagram (only two-input NAND, i.e., AND + NOT, gates may be used).
 - Implement the logic circuit on the breadboard with output connected to an LED and take a photo of your implementation. (only two-input NAND, i.e., AND + NOT, gates may be used).
 - Photo the output of each possible input pattern.

LAB 4-2

- Input: A, B, C ; Output: $F(A, B, C) = \Sigma(1, 5, 6, 7)$
- Requirements
 - Complete the truth tables in voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the derived simplest Boolean expression, draw the circuit diagram (only two-input NOR, i.e., OR + NOT, gates may be used).
 - Implement the logic circuit on the breadboard with output connected to an LED and take a photo of your implementation. (only two-input NOR, i.e., OR + NOT, gates may be used).
 - Photo the output of each possible input pattern.

LAB 4-3

- Input: A, B, C, D ; Output: $F(A, B, C, D) = \Sigma(0, 2, 4, 5, 8, 9, 10, 11, 12, 13)$
- Requirements
 - Draw the truth tables in voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the derived simplest Boolean expression, draw the circuit diagram (only two-input NAND, i.e., AND + NOT, gates may be used).
 - Implement the logic circuit on the breadboard with output connected to an LED and take a photo of your implementation. (only two-input NAND, i.e., AND + NOT, gates may be used).
 - Photo the output of each possible input pattern.

LAB 4-4

- Input: A, B, C, D ; Output: $F(A, B, C, D) = \Sigma(0, 2, 4, 5, 8, 9, 10, 11, 12, 13)$
- Requirements
 - Draw the truth tables in voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the derived simplest Boolean expression, draw the circuit diagram (only two-input NOR, i.e., OR + NOT, gates may be used).
 - Implement the logic circuit on the breadboard with output connected to an LED and take a photo of your implementation. (only two-input NOR, i.e., OR + NOT, gates may be used).
 - Photo the output of each possible input pattern.

Bonus

- $F = CD(B + C)A + (BC' + DE')$
- Requirements
 - Draw the truth tables in voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the simplest Boolean expression, draw the circuit diagram (**only two-input AND + NOT, i.e., NAND, gates can be used**).
 - Implement the logic circuit on the breadboard with output connected to an LED and take a photo of your implementation. (**only two-input AND + NOT, i.e., NAND, gates can be used**).
 - Photo the output of each possible input pattern.